



NAME 

SCHOOL 

TEACHER 

Pre-Junior Certificate Examination, 2019

Mathematics

Paper 1

Higher Level

Time: 2 hours, 30 minutes

300 marks

School Stamp

Running Total	
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For Examiner			
Question	Mark	Question	Mark
1		11	
2			
3			
4			
5			
6			
7			
8			
9			
10		Total	

Grade

Instructions

There are 11 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

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Question 1

(Suggested maximum time: 10 minutes)

- (a) Study the list of five numbers below.

$$3 \qquad \frac{7}{3} \qquad \pi \qquad 3.33 \qquad \sqrt{11}$$

Answer the following questions using **only** the numbers from this list.

- (i) Place the five numbers **in order**, starting with the smallest.

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- (ii) Write down an integer.

Answer =

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- (iii) Write down a rational number.

Answer =

--

- (iv) Write down an irrational number.

Answer =

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- (b) In Ireland, the population was estimated to be 4 819 308 on 12th November 2018.

- (i) The population was estimated to be 4 804 113 on 12th July 2018.
Calculate the percentage increase in the population over this four month period.
Give your answer correct to one decimal place.

- (ii) Ireland's population is equivalent to 0.06% of the total world population.
Find the total world population on 12th November 2018 in the form $a \times 10^n$,
where $n \in \mathbb{Z}$ and $1 \leq a < 10$, $a \in \mathbb{R}$, correct to three significant figures.

Question 2 (Suggested maximum time: 15 minutes)

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A sum of €30 000 was invested in a savings account, offering an annual compound interest rate of 2%. Tax of 35% will be deducted each year from the interest earned.

- (a)** Calculate the interest earned at the end of the first year before tax is deducted.

[illegible]

- (b)** Calculate the value of the investment at the end of the second year after tax is deducted.

A full-page view of a blank sheet of graph paper. The grid consists of small squares formed by thin gray lines. There are 20 columns and 15 rows of squares. The margins are uniform on all sides.

Another savings account guarantees that the same initial investment will be worth €32 000 at the end of two years, before tax is deducted.

- (c)** Find the annual compound rate of interest for this savings account. Give your answer correct to two decimal places.

A blank sheet of graph paper with a grid pattern. The grid consists of small squares formed by thin gray lines. There are 20 columns and 15 rows of squares. The entire sheet is covered by this grid, with no margins or other markings.

Question 3 (Suggested maximum time: 10 minutes)

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(a) Factorise fully each of the following expressions.

(i) $9x^2 - 81$

[illegible]

(ii) $3x^2 - 17x + 10$

[illegible]

(iii) $4a - 6bc + 3ac - 8b$.

[illegible]

(b) (i) Write the following as a single fraction in its simplest form.

$$\frac{x+2}{3} - \frac{x-1}{2}$$

[illegible]

(ii) Hence, or otherwise, solve the equation

$$\frac{x+2}{3} - \frac{x-1}{2} = 6.$$

[illegible]

Question 4

(Suggested maximum time: 15 minutes)

90 students were asked whether they had ever been to three of Ireland's most popular tourist attractions: Newgrange (N), the Cliffs of Moher (C) and the Giant's Causeway (G).

38 students had been to Newgrange.

49 students had been to the Cliffs of Moher.

7 students had been to the Giant's Causeway only.

5 students had been to all three of the attractions listed.

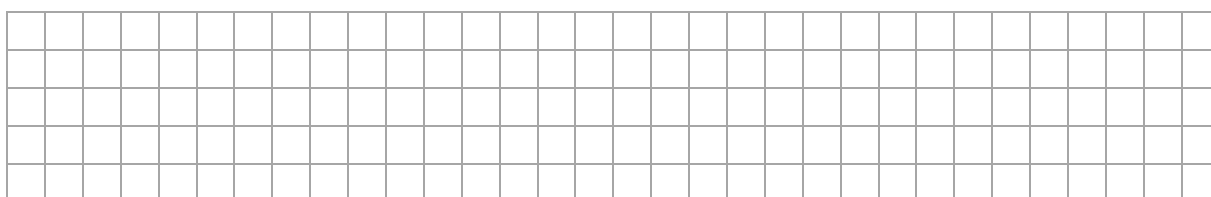
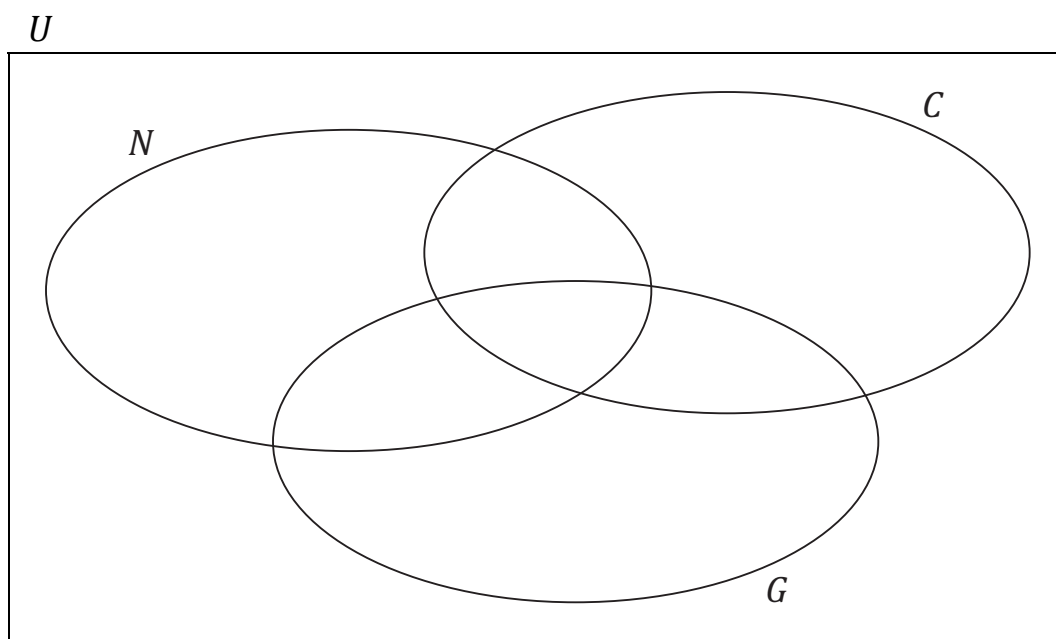
10 students hadn't been to any of the attractions listed.

12 students had been to the Giant's Causeway and Newgrange.

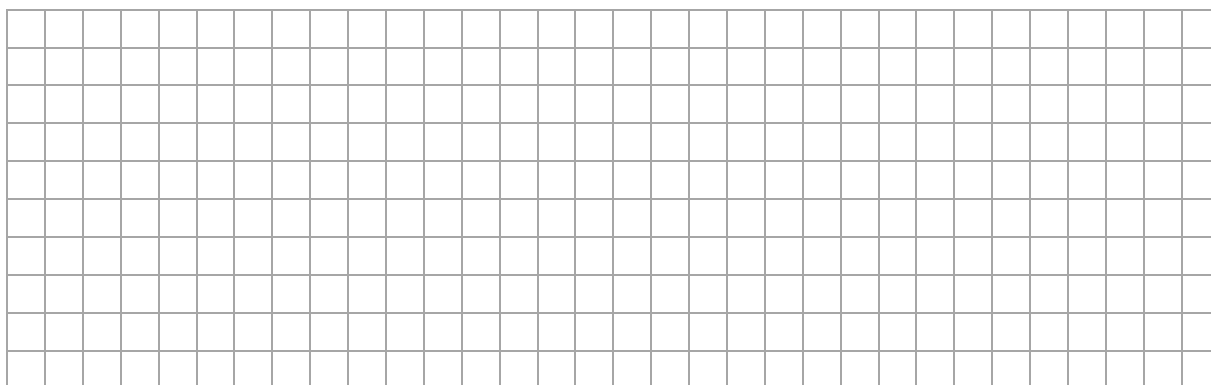
14 students had been to the Giant's Causeway and the Cliffs of Moher.

x students had been to the Cliffs of Moher and Newgrange but **not** the Giant's Causeway.

- (a) Use this information to fill in the Venn diagram below, in terms of x .



- (b) Find the value of x .



- (c) Find the **probability** that a student chosen at random from the group had been to the Giant's Causeway.

[illegible]

- (d) Find the **probability** that a student chosen at random from the group had been to **at least one** of the tourist attractions.

[illegible]

- (e) Find the **probability** that a student chosen at random from the group had been to **only two** of the tourist attractions.

[illegible]

- (f)** What type of data was collected during the survey? Tick (✓) **two** boxes only.

Numerical	
Categorical	

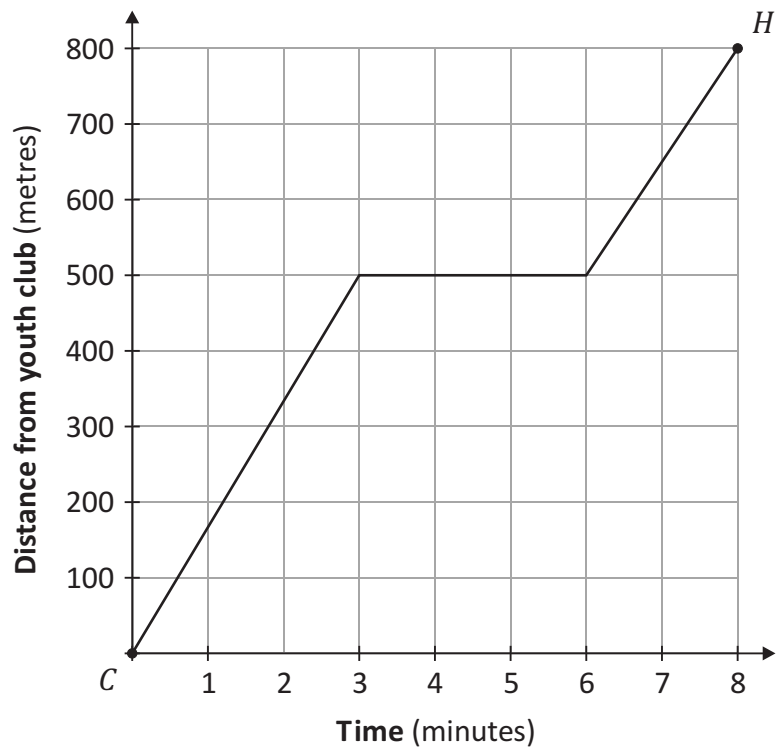
Nominal	
Ordinal	

Discrete	
Continuous	

Question 5

(Suggested maximum time: 15 minutes)

Twin brothers Frank and Walter are at their local youth club.
Frank leaves the club (C) at 7:00 p.m. and walks home (H), a distance of 800 metres.
The distance-time graph below shows his journey.



(a) Use the graph to estimate how far Frank had walked 2 minutes after leaving the youth club.

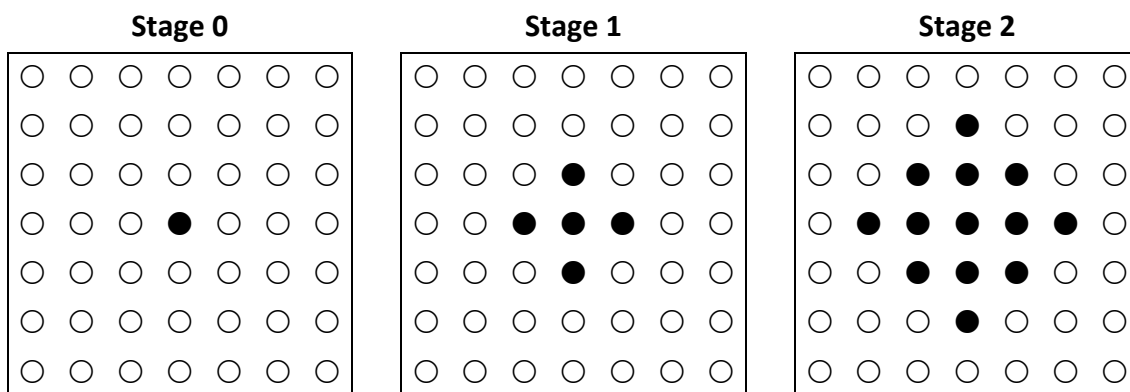
Distance Frank had walked = metres

(b) Only one sentence below matches Frank’s journey.
Place a tick (✓) in the box beside the correct sentence.

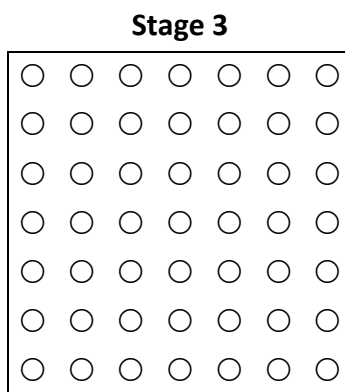
Statement	Tick one box only
Frank walks at a constant pace for 3 minutes, stops for 2 minutes to chat to a friend, then continues walking at a faster pace, arriving home at 7:08 p.m.	
Frank walks at a constant pace for 3 minutes, stops for 3 minutes to chat to a friend, then continues walking at the same pace, arriving home at 7:08 p.m.	
Frank walks at a constant pace for 3 minutes, stops for 3 minutes to chat to a friend, then continues walking at a faster pace, arriving home at 7:08 p.m.	
Frank walks at a constant pace for 3 minutes, stops to return to the youth club because he forgot his bag, then sprints home, arriving at 7:08 p.m.	
Frank walks at a constant pace for 3 minutes, stops for 3 minutes to chat to a friend, then continues walking at a slower pace, arriving home at 7:08 p.m.	

(Suggested maximum time: 10 minutes)

A gardening club is making a design for a walkway by arranging groups of black tiles in a pattern. In each stage the black tiles increase by adding a tile next to each of the black tiles inserted in the previous stage, both horizontally and vertically, but not diagonally. The first three stages of the sequence are shown below.



- (a)** Draw the next stage of the sequence in the design.



- (b)** Complete the table below to show the number of black tiles in each stage of the sequence.

Stage	0	1	2	3	4	5
Number of black tiles						

[illegible]

- (c) Put a tick in the correct box to show what kind of pattern is made by the black tiles in the sequence. Tick (✓) **one** box only. **Justify** your answer.

linear quadratic exponential none of these

☐ ☐ ☐ ☐

[illegible]

Question 7 (Suggested maximum time: 10 minutes)

Question 7 (Suggested maximum time: 10 minutes)

The following Sunday she worked 8 hours, earning €166.70 including tips.

Caroline received €15 more in tips on Sunday than Saturday.

- (a) Write an expression in h and t for Caroline's total earnings on each day, given that h is her hourly rate and t is the value of tips earned.

Saturday's expression	Sunday's expression

[illegible]

- (b)** Using your expressions from part **(a)**, or otherwise, find Caroline's hourly rate and the value of tips she earned each day.

[illegible]

Hourly rate:

€

Saturday tips:

€

Sunday tips:

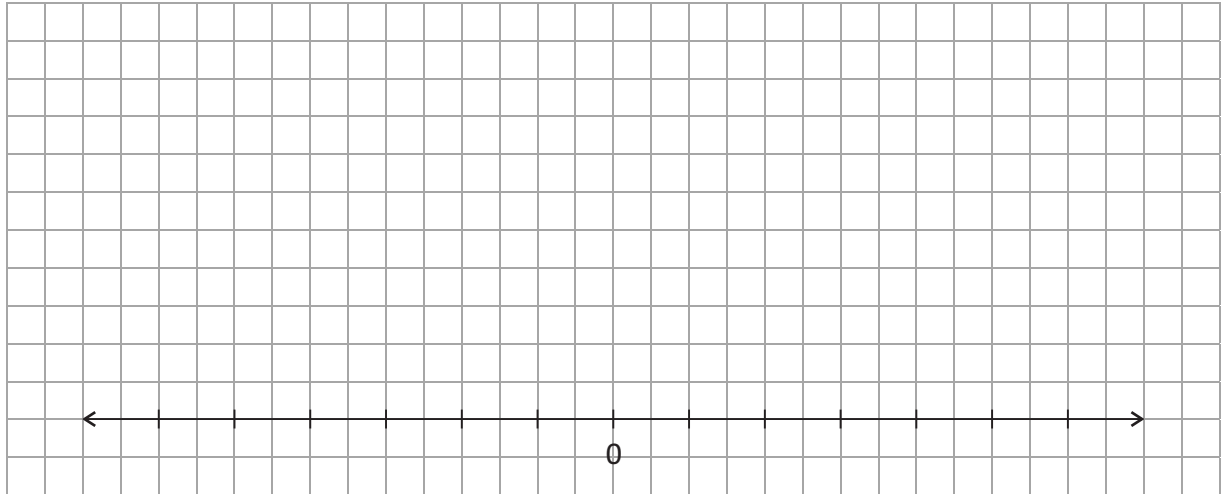
€

Question 8

(Suggested maximum time: 15 minutes)

- (a) Solve the following inequality and graph your solution on the number line.

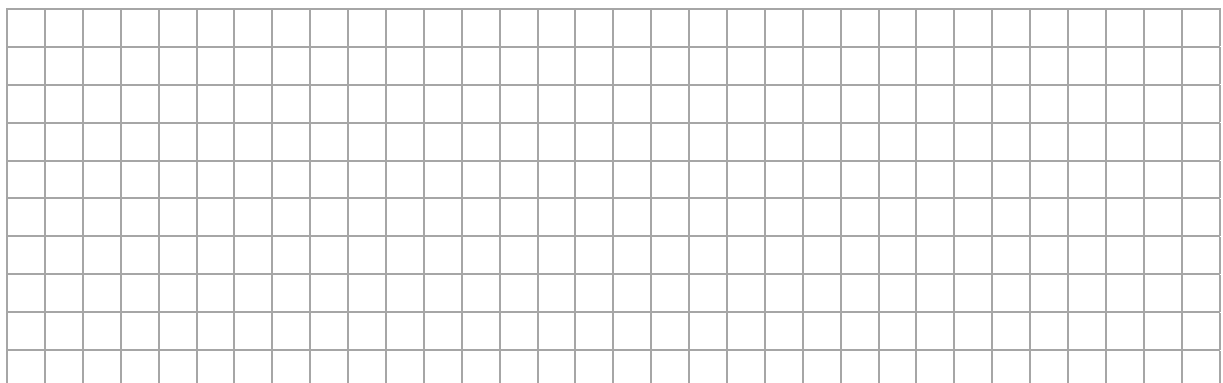
$$-4 < 2x - 1 \leq 9, x \in \mathbb{Z}.$$



- (b) Write an inequality in x , $x \in \mathbb{R}$, represented by each of the number lines shown below. The first one is done.

Number line	Inequality in x
	$0 < x \leq 4$

- (c) Multiply out and simplify $(3x + y - 4)(2y - 5)$.



Question 9

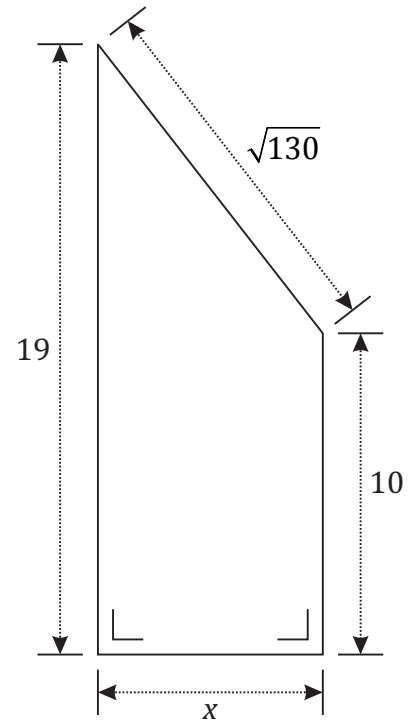
(Suggested maximum time: 15 minutes)

The diagram shown is a trapezoid.

- (a)** Write down the lengths of the parallel lines in the diagram.

[illegible]

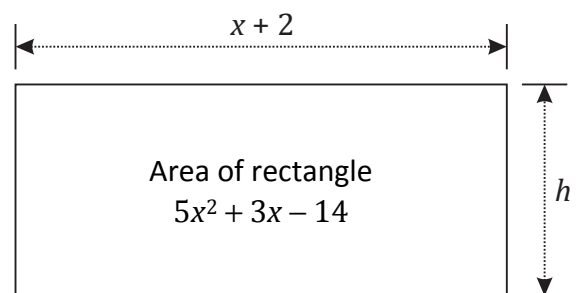
- (b)** Calculate the value of x .

[illegible]

- (c)** Find the area of the trapezoid.

[illegible]

- (d) The **area** of the rectangle shown is $5x^2 + 3x - 14$ square units. It has a width of length $x + 2$ units. Find the length of the side marked ***h*** in the diagram.



The graph shows a parabola opening upwards on a coordinate plane. The x-axis and y-axis are both labeled from -10 to 10 with grid lines every 1 unit. The parabola has its vertex at $(-0.5, -14.75)$ and passes through the points $(-2, -10)$ and $(3, -10)$. The equation $f(x) = 3x^2 + 3x - 14$ is written in the top right corner of the grid.

Question 10

(Suggested maximum time: 15 minutes)

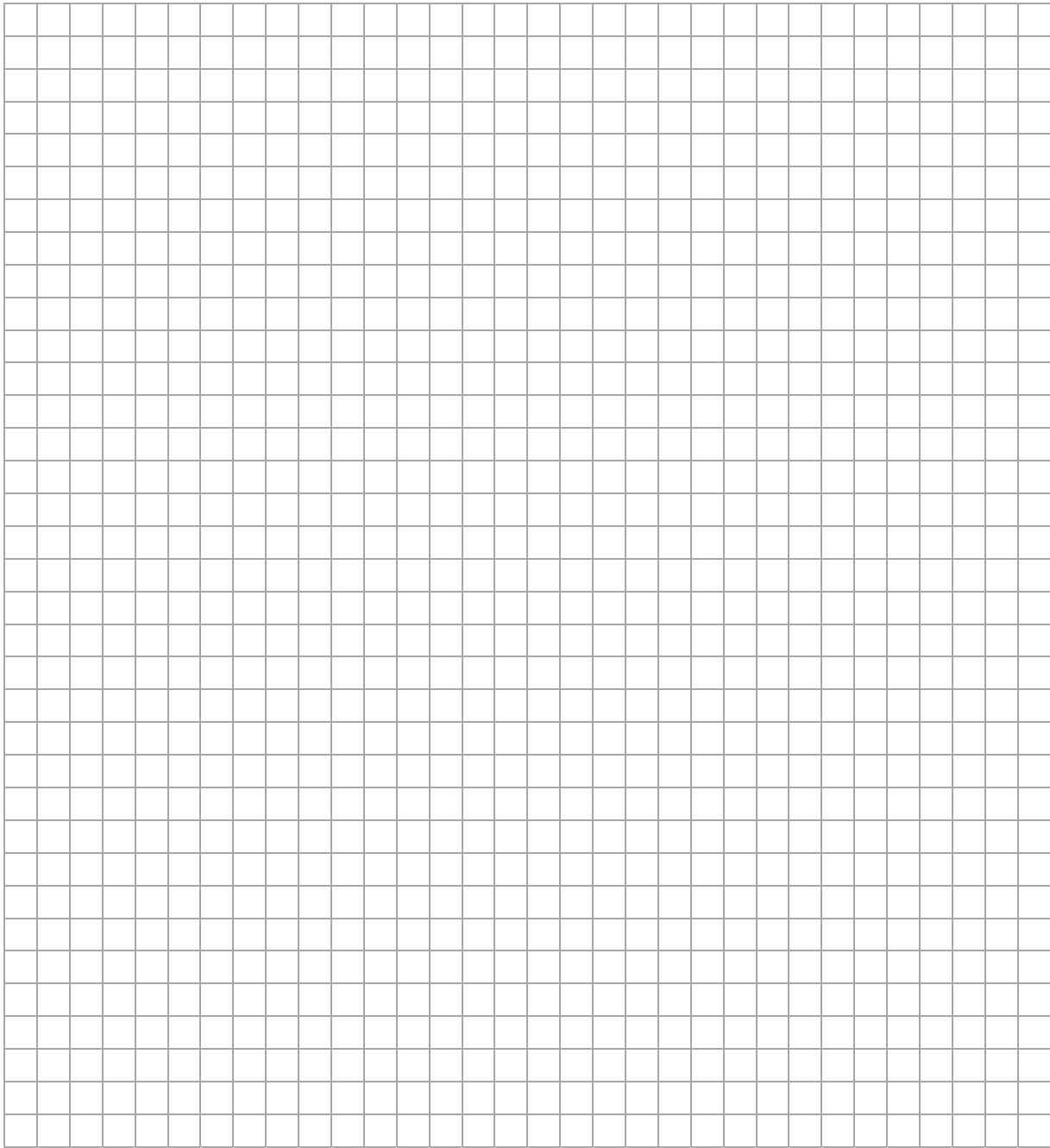
f is the function $f: x \mapsto 7 + 5x - 2x^2$, where $x \in \mathbb{R}$.

- (a)** Complete the table below by finding the values of $f(x)$ for $-1 \leq x \leq 4$, where $x \in \mathbb{R}$.

x	$f(x)$
-1	
0	
1	
2	
3	
4	

[illegible]

(b) Draw the graph of $f(x)$ in the domain $-1 \leq x \leq 4$, where $x \in \mathbb{R}$.



(c) Using the graph, write down the maximum value of $f(x)$.

(d) Write down the range of $f(x)$ in the domain $-1 \leq x \leq 4$.

Question 11

(Suggested maximum time: 10 minutes)

- (a)** Only one of the arrow diagrams A and B shown in the table below is a function.

Arrow diagram A	Arrow diagram B
<pre> graph LR subgraph S1 0 2 5 6 end subgraph S2 W X Y Z end 0 --> W 0 --> X 2 --> X 5 --> Y 6 --> Z </pre>	<pre> graph LR subgraph S1 A B C D end subgraph S2 1 3 9 11 end A --> 1 B --> 3 C --> 3 D --> 11 </pre>

Which arrow diagram is a function? Give a reason for your answer.

[illegible]

- (b)** Let $f(x) = 5x + 10$, where $x \in \mathbb{R}$.

- (i) Find the value of $f(4)$.

[illegible]

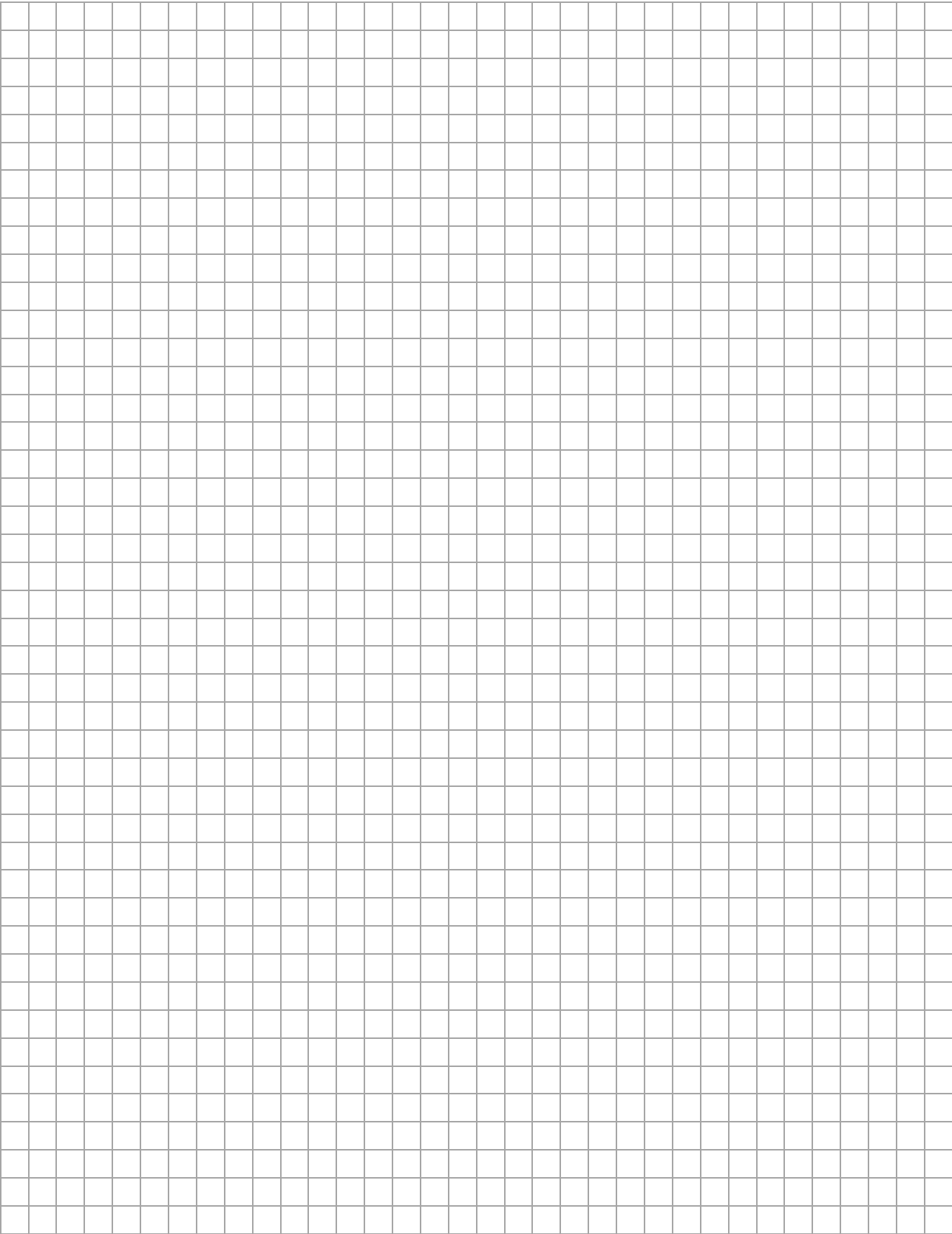
- (ii) Find the value of $f(k^2 + 1)$ in terms of k .

[illegible]

- (iii)** Find the value(s) of k for which $f(k^2 + 1) = 20$.

[illegible]

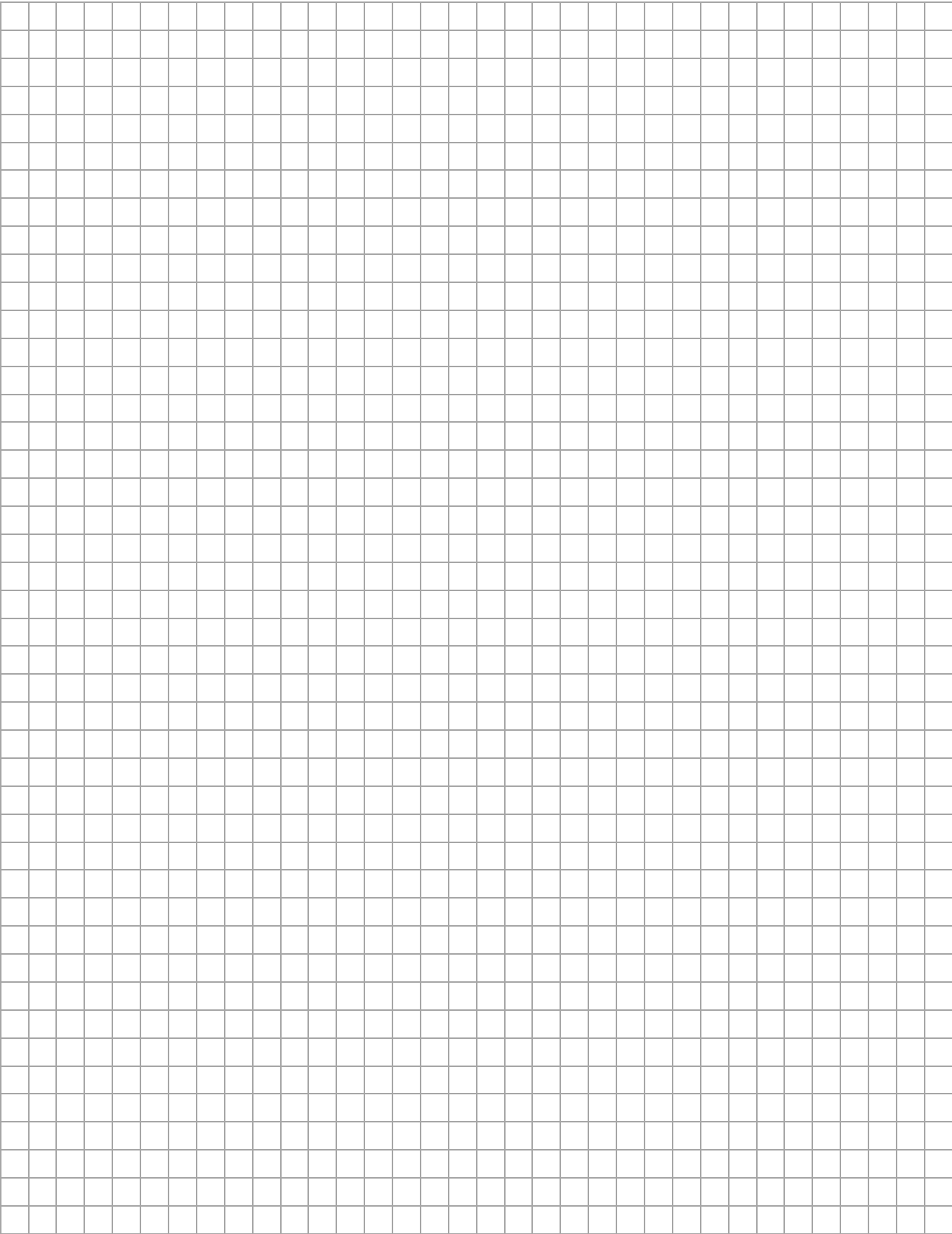
Page for extra work.
Label any extra work clearly with the question number and part.



Label any extra work clearly with the question number and part.

This image shows a full page of blank graph paper. The grid consists of thin, light gray horizontal and vertical lines that intersect to form small squares across the entire surface. There are no margins, text, or other markings on the paper.

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This image shows a full page of blank graph paper. The grid consists of small, uniform squares formed by thin, light gray lines. There are no margins, text, or other markings on the page.

Pre-Junior Certificate Examination, 2019

Mathematics – Paper 1

Higher Level

Time: 2 hours, 30 minutes

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